THE PULLBACK OF AN INVERTIBLE MODULE IS INVERTIBLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterisation of invertible modules as those admitting a module whose tensor product with them is the structure sheaf. Invoked twice in the counted proof (source lines 4367 and 4373) as a pointer to a lemma of this chapter. Supplies the inverse of the given module and tests invertibility of its pullback.
- The compatibility of the pullback with the tensor product of modules. Invoked inside the counted proof at source lines 4370-4372 as a pointer to a lemma of this chapter. Transports the trivialising isomorphism along the pullback.

Source reference. Original lemma lemma-invertible: sources/stacks-modules/pinned-original-source.tex, source lines 4367–4367. Navigation pointer only: the counted text refers to the Stacks item lemma-invertible, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-tensor-product-pullback: sources/stacks-modules/pinned-original-source.tex, source lines 4372–4372. Navigation pointer only: the counted text refers to the Stacks item lemma-tensor-product-pullback, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The pullback $f^*\mathcal{L}$ of an invertible $\mathcal{O}_Y$-module is invertible.*

Proof. By Lemma lemma-invertible there exists an $\mathcal{O}_Y$-module $\mathcal{N}$ such that $\mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{N} \cong \mathcal{O}_Y$. Pulling back we get $f^*\mathcal{L} \otimes_{\mathcal{O}_X} f^*\mathcal{N} \cong \mathcal{O}_X$ by Lemma lemma-tensor-product-pullback. Thus $f^*\mathcal{L}$ is invertible by Lemma lemma-invertible. □